Numerical Problem

IG (Information Gain)

* Decision Tree: ID3 Algorithm

Day	Weather	Temp	Humidity	Wind	Play
Day1	Sunny	Hot	High	Weak	NO
Day2	Sunny	Hot	High	Strong	NO
Day3	Cloudy	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	NO
Day7	Cloudy	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	NO
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Cloudy	Mild	High	Strong	Yes
Day13	Cloudy	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	NO

Calculate IG of weather

Step1: Entrophy of entire dataset

$$S\{+9,-5\} = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

Step2: Entrophy of all attributes

Entrophy of Sunny $\{+2,-3\} = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$

" of Cloudy $\{+4,-0\} = -\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4} = 0$

" Rain $\{+3,-2\} = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$

Information Gain $= \text{Entropy(whole data)} - \frac{5}{14}\,\text{Ent}(S)$

$- \frac{4}{14}\,\text{Ent}(C) - \frac{5}{14}\,\text{Ent}(R)$

$0.94 - \frac{5}{14}(0.97) - \frac{4}{14}(0) - \frac{5}{14}(0.97) = 0.246$

Calculate IG of Temperature

Step1: Entrophy of entire dataset

$S\{+9,-5\} = \frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$

Step2: Entrophy of all attributes:

Entropy of Hot $\{+2,+2\} = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4} = 1.0$

Entropy of Mild $\{+4,-2\} = -\frac{4}{6}\log_2\frac{4}{6} - \frac{2}{6}\log_2\frac{2}{6} = 0.91$

Entropy of Cold $\{+3,-1\} = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} = 0.81$

Information Gain $= \text{Entrophy(whole data)} - \frac{4}{14}\,\text{Ent}(H)$

$- \frac{6}{14}\,\text{ent}(M) - \frac{4}{14}\,\text{ent}(C) = 0.029$

Calculate IG of Humidity

Step1: Entrophy of entire dataset

$S\{+9,-5\} = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$

Step2: Entrophy of all attributes

Entrophy of High $(+3,-4) = -\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7} = 0.98$

Entrophy of Normal $\{+6,-1\} = -\frac{6}{7}\log_2\frac{6}{7} - \frac{1}{7}\log_2\frac{1}{7} = 0.59$

Information Gain $= \text{Intropy(whole data)} - \frac{7}{14}\,\text{Ent}(H) \frac{7}{14}\,\text{Ent}(H)$

$= 0.15$

Calculate IG of wind

Step1: Entrophy of entire dataset

$S\{+9,-5\} = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$

Step2: Entropy of all attributes:

Entropy of Strong $\{+3,-3\} = -\frac{3}{6}\log_2\frac{3}{6} - \frac{3}{6}\log_2\frac{3}{6} = 1.0$

Entropy of weak $\{+6,-2\} = \frac{6}{8}\log_2\frac{6}{8} - \frac{2}{8}\log_2\frac{2}{8} = 0.81$

Information Gain $= \text{Entropy(whole data)} - \frac{6}{14}\,\text{Entry}(S) - \frac{8}{14}\,\text{Ent}(W)$

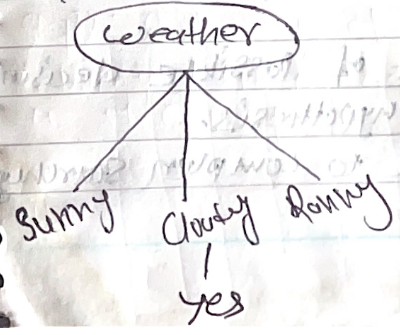
$= 0.0478$

Gain(S, weather) = 0.246

Gain(S, Temp) = 0.029

Gain(S, Humidity) = 0.15

Gain(S, wind) = 0.0478



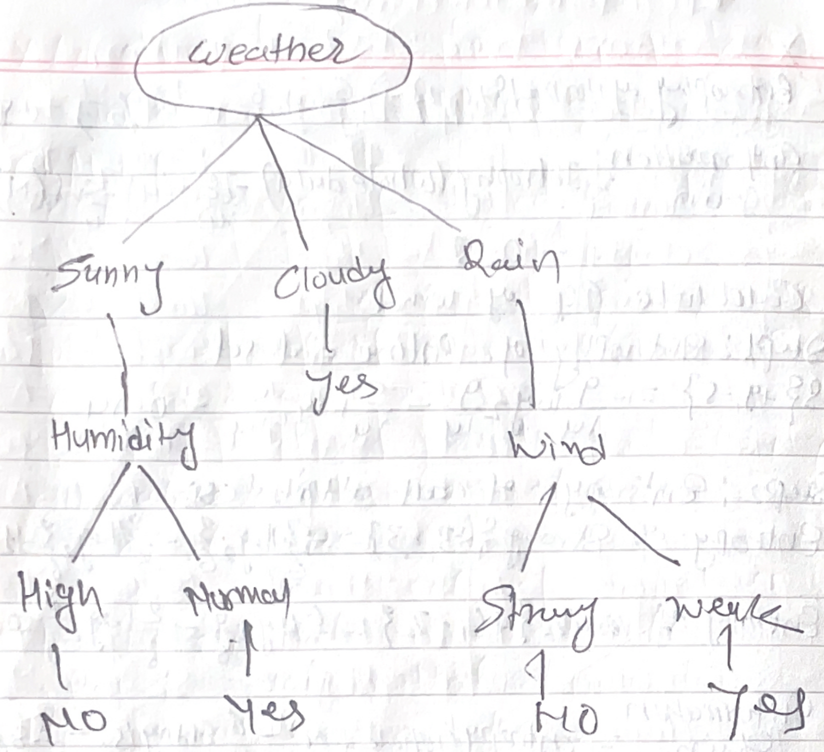
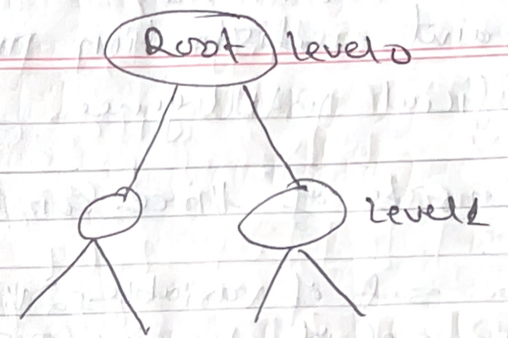


fig: ID3 algorithm (decision tree)

* Hypothesis space search in decision tree
→ ID3 can be characterised as searching a space of hypothesis for one that fits the training examples.

ID3: will search set of possible decision trees from available hypothesis.
ID3 performs simple to complex searching



first start with empty tree and keep on adding.
- every discrete valued (finite) function can be described by some decision tree
- avoids major risk of searching incomplete hypothesis
- has only single current hypothesis
- can not determine alternative decision tree
- Backtracking is not possible
- can be ~~extracted~~ extended easily to noisy data also

[Sudoku, map coloring, graph coloring
Cript-arithmetic problem
{used CSP method}]

* Constraint satisfaction problem in artificial intelligence

CSP

CSP consists of three components
V, D, C

V – is a set of variables $\{V_1, V_2 \ldots V_n\}$

D – is a set of domains $\{D_1, D_2 \ldots D_n\}$ one for each variables

c – is set of constraints that specify allowable combination of values.

$C_i = (\text{Scope}, \text{rel})$

→ where scope is set of variables that participate in constraints

→ Rel is relation that define the values that variable can take

$V_1\ V_2$ $C_1 = \{V_1, V_2\}$

A B $V_1 \neq V_2$

(1,2) (2,4)

$C_1 = (V_1, V_2), (A, B)$

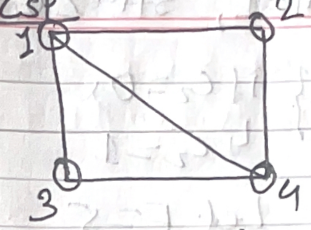
$C_1 = \{(V_1, V_2)\ (1,2), (1,4)\ (2,4)\}$

1	2	3	4	5	9x9
2	3	4	5	6	87
3	4	5	6	7	(1+9)
4	5	6	7	8	
5	6	7	8	9	

3+3

Constant graph

CSP



$V = \{1, 2, 3, 4\}$

$D = \{$Red, Green, Blue$\}$

$C = \{1 \neq 2, 1 \neq 3, 1 \neq 4, 2 \neq 4, 3 \neq 4\}$

Backtracking algo.

Intelligent

	1	2	3	4
Initial Domain	RGB	RGB	RGB	RGB
1 = R	R	GB	GB	GB
2 = G	R	G	GB	B
3 = B	R	G	B	B

empty

Intelligent backtrack

3 = G	R	G	G	B

Crypt-arithmetic problem

↳ type of constraint satisfaction problem

↳ constraints → No two letters have same value

There should be only one carry forward.

Sum of digit must be as shown in problem

Digits that can be assigned to a word/alphabet (0–9) range.

```
  TO
+ GO
----
 OUT
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Left	Dig
T	2
O	1
G	8
U	0

LMd – 1
≠ 0
0 – 9
9 + 8
17
carry

$2 + G = U + 10$

$2 + 9 = 11$

$2 + 8 = 10$

2	1
8	1

1 0 2